

Phase 02 - Sputtering-Annealing

Explanation and Operating Logic - Unified GUI

Release: 0.9.36 **Build:** 2026.08.11-r15

Operator and developer explanation of the current Phase 02 task-focused GUI, automated COSCON workflow, manual leak-valve actions, and safety behavior.

1. Role in the synthesis workflow

- Run repeated sputtering and annealing cycles, normally three cycles.
- Automate COSCON Degas/Operate/Standby sequencing through direct UDP control.
- Keep the argon leak valve operator-controlled while the software guides when to open and close it.
- Write and verify the oven PID annealing setpoint and monitor pressure/COSCON telemetry continuously.

2. Unified launcher and GUI

Unified launcher: START_NPG_CHAMBER.bat opens the main launcher. Each phase starts as a separate child process and owns its serial ports and hardware cleanup.

Phase-local shutdown: Abort / Safe Stop closes only the active phase after its safe hardware action. The unified launcher remains open and can be used again after the child process releases its ports.

Launcher close protection: while a phase is active, the launcher itself refuses to force-kill it. Use the phase GUI Abort / Safe Stop or let the phase finish normally.

GUI area	Purpose
Workflow card	Shows the current cycle/stage and only the operator action that is currently valid.
Telemetry	Pressure, oven PV/SV, COSCON mode and measured output values.
Process sequence	Degas (optional), open valve, pressure conditioning, COSCON activation, sputter timer, Standby, close valve, anneal.
Abort / Safe Stop	Stops the active Phase 02 process through its own safe-stop path.

3. Default process

Parameter	Default
Cycles	3
Sputtering time	20 min per cycle
Annealing target	620 C
Annealing hold	10 min
PID reset after anneal	0 C
Pressure warning	5e-5 mbar
Pressure emergency limit	1e-4 mbar

Cycle 1 normally performs COSCON Degas. The launcher can explicitly start without initial Degas when continuing a previously degassed chamber preparation.

4. COSCON and pressure safety

- COSCON activation is blocked when pressure is unavailable or above the configured emergency limit.
- Operate is verified using measured energy/emission and interlock/device state.
- A specific transient activation Energy Overload can receive one controlled recovery attempt; repeated overload aborts activation.
- COSCON is returned to a verified inactive safe state before the operator closes the leak valve.

5. Abort behavior

Abort / Safe Stop is a phase-local stop. It attempts the configured Phase 02 hardware safe state, including oven PID reset to 0 C and COSCON safe state as applicable, then closes Phase 02 without closing the unified launcher.

6. Data and operator checklist

- Keep other COSCON control software closed while direct Phase 02 control is active.
- Open/close the argon leak valve only when the GUI requests that physical action.
- Watch pressure and COSCON measured output during sputtering.
- Do not continue after an overload, interlock change or emergency pressure event until the hardware state is understood.

Operational note: Software state is not a substitute for checking the real chamber hardware. Verify shutters, pressure, PID, Keysight output and any manually operated hardware before continuing.